

ASSIGNMENT 2.3

100Marks

PROBLEM STATEMENT

In a Distributed Hash Table (DHT) based file sharing system, digital content is stored and accessed across multiple independent nodes using a structured mapping. Unlike flooding where every node in the network is queried, a DHT uses a simple hash function to deterministically calculate exactly which node stores a specific file, significantly reducing network traffic and search times.

Implement a single-threaded, decentralized DHT file-sharing system in C. Each node in the system must maintain multiple TCP connections simultaneously. Instead of a flooding query, nodes must maintain a list of all connected peers and use a simple modulo-based hashing mechanism (e.g., **$\text{Hash}(\text{filename}) \% \text{Total_Connected_Peers}$**) to distribute and locate files across the network. The system must remain entirely responsive to user input and network events without spawning multiple threads or processes.



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- For each peer run the peer code in a different terminal.
- Create a folder containing files where each peers file are being stored.
- When a peer shares a file into the network, the system must hash the filename, identify the responsible peer from its known peer list, and transfer the file to that specific peer's folder.
- When a peer searches for a file, it must compute the hash to find the responsible peer and request the file directly. After receiving, the file should be present in the requesting peer's folder.
- Analyze how your chosen hash function affects the amount of content replication and distribution evenly (or unevenly) across the peers. Show it clearly.

